Evaluation of the mixture fits: recommendation, price counterfactuals and basket generation

Version-4 basket model, branch `v4-mixture`

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All numbers are on the *validation weeks* (`trip_split = 1`). These trips were never used in training or model selection. The test weeks were not read. Tool: `research/evaluate_mixture.py`. Its protocols are those of the evaluators removed in `d716f95`, so results compare with earlier ones. Only the treatment of the taste z changes, to the mixture form.

1 What is evaluated

Checkpoints. Real data (Dunnhumby, 5,455 products):

- **step 360**: the old best mixture checkpoint.
- **R2+fw**: the new recipe. It continues from step 360 to step 840 without whitening, then applies one final compensated whitening (`docs/training_theory.pdf`).
- **R0**: the old recipe continued to step 840.

Synthetic world `disc_c48_s3` (522 products, with causal truth for price changes):

- the control's best (step 420) and final (step 1,210) checkpoints;
- S1: household and product-context blocks frozen;
- S9: S1 with Adam's state kept across whitening, the best synthetic fit.

Recommendation. One product is hidden from each basket that has at least 2 products. Every product the store offers is then ranked by $\log p(\text{rest} + j) - \log p(\text{rest})$. Because Z_z is the same for both baskets, this equals

$$\log \mathbb{E}\big[\exp \Delta E_z(j) \,\big|\, \text{rest}\big], \qquad \Delta E_z(j) = d_j - s \cdot \Delta s_j - \tfrac{1}{2}|\Delta s_j|^2 + z \cdot \Delta s_j.$$

Here d_j is the exact change of the energy, and Δs_j the change of the basket field. The expectation is over the posterior of z given the rest, with 32 importance draws. The cases are 2,000 validation trips, with a fixed seed. The table reports MRR, recall@k and the median rank; standard errors are clustered by household. "No taste integral" ranks by d_j alone, which is the old form's formula.

Generation. One basket per validation trip: $z \sim \mathcal{N}(0, I)$, then $S \mid z$ exactly. A gradient-boosted classifier tries to tell real from generated baskets, using the full basket or only the categories present. AUC 0.5 means the two cannot be told apart. Three seeds; 6,000 trips on real data, and all 2,755 on synthetic data.

Price counterfactuals. *Synthetic:* 32 scenarios, each multiplying one product's price by 0.8 or 1.2, compared with the generator's causal truth. The quantities are the relative change in purchases of the product, of the rest of its subcategory and of the rest of its category, and the change in basket size. The model's incidence is the mean, over 64 common draws $z \sim \mathcal{N}(0, I)$ per trip, of the exact $P_z(j \in S)$, on 400 validation trips. *Real:* 166,164 validation-week discount events (a product's chain discount depth changes between weeks). The measured within-category response comes from a conditional binomial with an offset for the category's change, with product-clustered 95% intervals. The model's response is the depth slope, the display and mailer effects, and its agreement with the observed share changes. It uses 580 trips (one per store-week) and 64 draws of z.

2 Results

Recommendation (real: 1,620 cases; synthetic: 1,974).

	model	MRR	recall@10	recall@20	median rank
real	step 360	0.119 ± 0.007	0.228	0.305	92
	R2+fw	$\mathbf{0.123 \pm 0.007}$	0.227	0.306	86
	R0	0.116 ± 0.007	0.227	0.293	100
	no taste integral (R0)	0.076 ± 0.006	0.132	0.177	264
	popularity	0.045 ± 0.004	0.075	0.097	760
synthetic	control best	0.259 ± 0.008	0.485	0.613	12
	control final	0.277 ± 0.008	0.502	0.642	10
	S1	0.276 ± 0.008	0.509	0.648	10
	S9	$\mathbf{0.278 \pm 0.008}$	0.513	0.652	10
	popularity	0.084 ± 0.005	0.166	0.280	50

Paired differences, household-clustered:

- real: R2+fw − step 360 = $+0.0034 \pm 0.0024$; R0 − step 360 = -0.0037 ± 0.0025;
- synthetic: S9 − control best = $+0.019 \pm 0.004$.

The model is 2.7 times better than popularity on real data and 3.3 times on synthetic data. Integrating over the taste of the rest of the basket matters: on real data, MRR is 0.116 with it and 0.076 with the old form's formula. On real data the differences between the three checkpoints are small, and only the ordering (new > old best > old recipe continued) is consistent with the likelihood.

Generation.

	model	full AUC	category AUC	size (real 7.50 / 8.88)	categories (5.88 / 8.11)	20+ products
real	step 360	0.709	0.661	8.92	7.42	0.130 (real 0.092)
	R2+fw	**0.675**	**0.633**	**7.24**	**5.96**	**0.080**
	R0	0.711	0.682	8.62	7.07	0.119
synthetic	control best	0.765	0.844	8.72	7.95	
	control final	0.687	0.788	7.81	7.19	
	S1	0.686	0.786	8.01	7.37	
	S9	**0.683**	**0.786**	7.98	7.35	

On real data the new recipe gives the most realistic baskets. Its AUC falls by 0.034 (full) and 0.028 (categories) against step 360. Its size is within 3.5% of the data, against 19% too large for step 360. The number of categories is within 1.4%, against 26% too many. On synthetic data, the later checkpoints are much more realistic than the control's best, even though the control's held-out likelihood fell after step 420. Their baskets are 10% too small, the shape gap of `training_theory.pdf`, Section 6.

Price counterfactuals, synthetic (32 scenarios; S9; the control is within 0.02 on every number).

	correlation with truth	sign agreement	slope of model on truth	mean change at ×0.8 / ×1.2
own product	0.93	1.00	0.10	truth $+0.47/-0.35$, model $+0.05/-0.04$
rest of subcategory	0.83	0.97	0.06	
rest of category	0.92	0.95	0.13	
basket size	−0.19	0.56		truth and model both ≈ 0

The directions and the ranking of the scenarios are right; the sizes are about ten times too small. A 20% price cut raises the product's purchases by 47% in the truth and by 5% in the model, so the own-price elasticity is about −2 in the truth and −0.25 in the model. Substitution to the rest of the subcategory and the category is also 6–13% of the truth. This holds for every synthetic checkpoint. In S1 and S9 the price parameters (γ_h, β) were not trained by the mixture fit, and the control is no better, so the gap comes from the base model's price terms, not from the mixture training.

Price and promotion response, real (validation weeks; all three checkpoints within 0.005 of each other, because the real fits did not train the price and promotion terms).

	measured (product-clustered 95%)	model
discount depth, within-category share slope	0.64 [0.35, 0.93]	1.38 (whole week) – 1.50 (own price only)
display on vs off, share log ratio	0.375 [0.28, 0.47]	0.37 (purchase-weighted), 0.52 (mean)
mailer on vs off	0.347 [0.26, 0.43]	0.39 (purchase-weighted), 0.51 (mean)
agreement with observed share changes		correlation 0.16, sign agreement 0.61
mean share change when the discount deepens	+0.108	+0.015

Display and mailer effects match the measured ones, weighted by purchases. The response to the discount depth is about twice the measured one, and outside its interval. Agreement event by event is weak, as before (`v4_comp_promo`: 1.38–1.51, correlation 0.15–0.16, the same).

3 Verdict

1. **Recommendation:** good, and the new recipe is the best on real data. The model is 2.7–3.3 times better than popularity, and the taste integral is worth 50% of the MRR on real data.

2. **Basket generation:** the new recipe is clearly the most realistic on real data: lowest classifier AUC, size within 3.5%, categories within 1.4%. A classifier can still tell real from generated baskets (AUC 0.68 full, 0.63 categories).

3. **Price counterfactuals: not good enough yet.** On synthetic data the signs and the ranking are right, but the sizes are about 10 times too small. On real data the discount response is twice the measured one, while display and mailer match. In neither case does the mixture training cause this. It sits in the price terms of the base model, which the mixture fits keep fixed (real) or near their start (synthetic). This is the next problem to work on. The goal requires correct price counterfactuals, and none of today's fixes touched them.

4 The price fix (added after the evaluation)

Diagnosis, synthetic. A within-category conditional logit on the synthetic training weeks (73,633 choices; product fixed effects) estimates −2.01 per unit of log regular price and −1.55 per unit of log discount, with 0.08 and 0.07 standard deviations of the two. So the data identify a large regular-price effect. The model's regular-price coefficient was $g_{hj}\kappa \approx 0.29$, against a discount coefficient of 1.73, and the scenarios change the regular price. There were two causes.

1. κ was limited by its learning rate. Adam moves a parameter by at most about its learning rate per step, so over 1,210 steps at 0.003 the raw κ can move at most 3.6. It moved 3.1. With a learning rate of 0.05 or 0.015 (runs S10, S11), κ converges at 11–12, and the response doubles (slope on truth 0.10 to 0.21).

2. The rest is the parametrisation. $g_{hj} = \text{softplus}(\gamma_h) \cdot \text{softplus}(\beta_j)$ is fixed at the base model's values (and sits where softplus is flat). So κ can only rescale a pattern that does not match the true sensitivities, and the best scale for it is shrunk. The discount part did not suffer from this because it has a common slope.

Fix. A common regular-price slope, $-c\,(\text{dlp} - m)$, so the regular-price coefficient is $g_{hj}\kappa + c$ (`v4/model.py`, `price_common`; `fit_mixture.py --price-common`), with the price scalars at learning rate 0.05. The law stays exact: it is one more term in each product's utility.

Result (run S12 = S9 + the fix). $c = 2.12$, $\kappa = 3.59$: the regular-price coefficient is 2.30 and the discount's about 2.2, against the data logit's 2.0 and 1.5.

	S9	S10 (κ converged)	**S12 (common slope)**	truth
own: slope of model on truth	0.10	0.21	**0.73**	1
own: mean change at $\times 0.8$ / $\times 1.2$	$+0.05/-0.04$	$+0.11/-0.08$	$+\mathbf{0.42}/-\mathbf{0.25}$	$+0.47/-0.35$
own: mean absolute error	0.36	0.31	**0.14**	
rest of category: slope on truth	0.13	0.28	**0.93**	1
same subcategory: slope on truth	0.06	0.12	0.44	1
held-out $\log p$ (1,024 trips)	-25.79		$-\mathbf{25.74}$ ($+0.055 \pm 0.010$)	
recommendation MRR	0.278		0.277 (-0.001 ± 0.002)	
generation AUC, full / categories	0.683 / 0.786		0.681 / 0.783	

The likelihood improves, and recommendation and generation are unchanged. Two parts of the generator are not in the model, and they explain the rest: price rises hurt twice as much as cuts (loss aversion), and products are nested by subcategory. So rises and within-subcategory switching are still understated.

Real data. The same logit on 152,298 real training-week choices gives -0.18 for regular price and -1.01 for discount depth (display 0.63, mailer 0.53). The real model's coefficients already agree with it: regular median 0.07 and mean 0.24, discount 0.91. So on real data the $2\times$ gap is *not* this problem. It is a gap between the choice logit, which pools all weeks (≈ 1.0), and the week-to-week discount-event study (0.64), which nets out pantry loading and post-promotion dips. Closing it needs a decision: which of the two the model should reproduce, and whether to add purchase dynamics (stock at home). It is not a training fix.